

8a	force per unit charge force on positive charge	B1 B1
bi	four straight vertical parallel lines, approximately evenly spaced arrows downwards	B1 B1
ii	$E = \frac{V}{d}$ $E = \frac{430}{0.067}$ $= 6.4 \times 10^3$ N C^{-1}	C1 A1
iii	smooth curve within plates and straight lines outside plates direction of deflection shown as upwards	B1 B1
ci	into the page	B1
ii	forces are in opposite directions and are equal	B1
Iii	$Eq = Bqv$ $B = \frac{E}{v} = \frac{6.4 \times 10^3}{2.6 \times 10^7}$ $= 2.5 \times 10^{-4}$ T	C1 A1
d	induced e.m.f. proportional to	B1

	rate of change of magnetic flux linkage	B1
e	Flux linkage $= BAN$ $= \pi \times 10^{-3} \times 2.8 \times \pi \times (1.6 \times 10^{-2})^2 \times 85$ $= 6.0 \times 10^{-4}$ Wb	B1
f	$\text{e.m.f.} = \frac{\Delta N\phi}{\Delta t}$ $= \frac{(6.0 \times 10^{-4} \times 2)}{0.30}$ $= 4.0$ mV	C1 A1
g	Sketch: $E = 0$ for $t = 0 \rightarrow 0.3$ s, 0.6 s $\rightarrow$ 1.0 s, 1.6 s $\rightarrow$ 2.0 s $E = 4$ mV for $t = 0.3$ s $\rightarrow$ 0.6 s (either polarity) $E = 2$ mV for $t = 1.0$ s $\rightarrow$ 1.6 s with opposite polarity	-1 per error